

ASSIGNMENT 1

IEDA 1250: OPTIMIZATION FOR DECISION MAKING (SUMMER, 2026)

1. (10 points) Solve the following linear programs (LPs) graphically. Note that the LPs may not always have optimal solutions. Report what you find.

(a)

$$\begin{array}{ll}\min_{x_1, x_2} & x_1 + x_2 \\ \text{s.t.} & x_1 + 2x_2 \leq 8 \\ & 3x_1 + 2x_2 \leq 12 \\ & x_1 + 3x_2 \geq 13\end{array}$$

Solution: This LP is unbounded

(b)

$$\begin{array}{ll}\min_{x_1, x_2} & 2x_1 - x_2 \\ \text{s.t.} & x_1 - x_2 \leq 1 \\ & 2x_1 + x_2 \geq 6\end{array}$$

Solution: This LP is unbounded

(c)

$$\begin{array}{ll}\min_{x_1, x_2} & x_1 + 3x_2 \\ \text{s.t.} & 5x_1 + 10x_2 \leq 50 \\ & x_1 + x_2 \geq 1 \\ & x_2 \leq 4 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$

Solution: This LP has a unique optimal solution ($x_1^* = 1, x_2^* = 0$)

(d)

$$\begin{array}{ll}\min_{x_1, x_2} & x_1 + 3x_2 \\ \text{s.t.} & x_1 + 3x_2 \geq 3 \\ & x_1 + x_2 \geq 2 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$

Solution: This LP has infinite optimal solutions. Every solution on $x_1 + 3x_3 = 3$ and between $(3/2, 1/2)$ and $(3, 0)$ is optimal.

2. (10 points) In class, we have discussed how to write the following primal problem

$$\begin{array}{ll}\max_{x_1, x_2} & 45x_1 + 80x_2 \\ \text{s.t.} & 5x_1 + 20x_2 \leq 400 \\ & 10x_1 + 45x_2 \leq 450 \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$

into its dual form

$$\begin{aligned}
& \min_{\mu_1, \mu_2} 400\mu_1 + 450\mu_2 \\
& s.t. \quad 5\mu_1 + 10\mu_2 \geq 45 \\
& \quad \quad 20\mu_1 + 45\mu_2 \geq 80 \\
& \quad \quad \mu_1 \geq 0, \mu_2 \geq 0.
\end{aligned}$$

In this question, we will introduce a second way of writing the dual, through which we can learn a lot of tricks in writing LP models.

Specifically, consider the primal LP:

$$\begin{aligned}
& \max_{x_1, x_2} 3x_1 + 4x_2 \\
& s.t. \quad -x_1 - 3x_2 = 5 \\
& \quad \quad 3x_1 - 5x_2 \geq 8 \\
& \quad \quad x_1 \leq 0
\end{aligned}$$

Then answer the following questions:

- (a) If we set $x_1 = -\bar{x}_1$ and $x_2 = \bar{x}_2 - \underline{x}_2$, where $\bar{x}_1 \geq 0$, $\bar{x}_2 \geq 0$, and $\underline{x}_2 \geq 0$ are our new decision variables. Can you substitute x_1 and x_2 and rewrite the LP in terms of $\bar{x}_1$, $\bar{x}_2$, and $\underline{x}_2$?

$$\begin{aligned}
& \max_{\bar{x}_1, \bar{x}_2, \underline{x}_2} -3\bar{x}_1 + 4\bar{x}_2 - 4\underline{x}_2 \\
& s.t. \quad \bar{x}_1 - 3\bar{x}_2 + 3\underline{x}_2 = 5 \\
& \quad \quad -3\bar{x}_1 - 5\bar{x}_2 + 5\underline{x}_2 \geq 8 \\
& \quad \quad \bar{x}_1 \geq 0, \bar{x}_2 \geq 0, \underline{x}_2 \geq 0
\end{aligned}$$

- (b) Based on the LP you obtained in (a), can you rewrite the equality constraint with a “ $\leq$ ” and a “ $\geq$ ” constraints?

$$\begin{aligned}
& \max_{\bar{x}_1, \bar{x}_2, \underline{x}_2} -3\bar{x}_1 + 4\bar{x}_2 - 4\underline{x}_2 \\
& s.t. \quad \bar{x}_1 - 3\bar{x}_2 + 3\underline{x}_2 \leq 5 \\
& \quad \quad \bar{x}_1 - 3\bar{x}_2 + 3\underline{x}_2 \geq 5 \\
& \quad \quad -3\bar{x}_1 - 5\bar{x}_2 + 5\underline{x}_2 \geq 8 \\
& \quad \quad \bar{x}_1 \geq 0, \bar{x}_2 \geq 0, \underline{x}_2 \geq 0
\end{aligned}$$

- (c) Based on the LP you obtained in (b), can you change the “ $\geq$ ” inequalities to “ $\leq$ ” inequalities?

$$\begin{aligned}
& \max_{\bar{x}_1, \bar{x}_2, \underline{x}_2} -3\bar{x}_1 + 4\bar{x}_2 - 4\underline{x}_2 \\
& s.t. \quad \bar{x}_1 - 3\bar{x}_2 + 3\underline{x}_2 \leq 5 \\
& \quad \quad -\bar{x}_1 + 3\bar{x}_2 - 3\underline{x}_2 \leq -5 \\
& \quad \quad 3\bar{x}_1 + 5\bar{x}_2 - 5\underline{x}_2 \leq -8 \\
& \quad \quad \bar{x}_1 \geq 0, \bar{x}_2 \geq 0, \underline{x}_2 \geq 0
\end{aligned}$$

- (d) Now the LP you obtained in (c) should only contain “ $\leq$ ” constraints and non-negative variables. Can you write the dual form of this LP? Please simplify the dual LP as much as possible.

$$\begin{aligned}
 \min_{\mu_1, \mu_2, \mu_3} \quad & 5\mu_1 - 5\mu_2 - 8\mu_3 \\
 \text{s.t.} \quad & \mu_1 - \mu_2 + 3\mu_3 \geq -3 \\
 & -3\mu_1 + 3\mu_2 + 5\mu_3 \geq 4 \\
 & 3\mu_1 - 3\mu_2 - 5\mu_3 \geq -4 \\
 & \mu_1 \geq 0, \mu_2 \geq 0, \mu_3 \geq 0
 \end{aligned}$$

which can be simplified as

$$\begin{aligned}
 \min_{\mu_0, \mu_3} \quad & 5\mu_0 - 8\mu_3 \\
 \text{s.t.} \quad & \mu_0 + 3\mu_3 \geq -3 \\
 & -3\mu_0 + 5\mu_3 = 4 \\
 & \mu_0 \text{ free}, \mu_3 \geq 0
 \end{aligned}$$

3. (10 points) Consider the following primal LP:

$$\begin{aligned}
 \min \quad & 2x_1 + 3x_2 + 5x_3 + 2x_4 + 3x_5 \\
 \text{s.t.} \quad & x_1 + x_2 + 2x_3 + x_4 + 3x_5 \geq 4, \\
 & 2x_1 - x_2 + 3x_3 + x_4 + x_5 \geq 3, \\
 & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0.
 \end{aligned}$$

Suppose that y_1 and y_2 are the dual variables that correspond to the first and second constraints in the primal LP, respectively, and that their optimal dual solution values are $y_1^* = 0.8$ and $y_2^* = 0.6$. Can you obtain the optimal primal solution using the complementary slackness condition? What is the optimal objective function value for both the primal and dual LPs?

Solution: The dual LP is

$$\begin{aligned}
 \max_{y_1, y_2} \quad & 4y_1 + 3y_2 \\
 \text{s.t.} \quad & y_1 + 2y_2 \leq 2 \\
 & y_1 - y_2 \leq 3 \\
 & 2y_1 + 3y_2 \leq 5 \\
 & y_1 + y_2 \leq 2 \\
 & 3y_1 + y_2 \leq 3 \\
 & y_1 \geq 0, y_2 \geq 0
 \end{aligned}$$

Using the optimal dual solution values, we have

$$\begin{aligned}
 y_1^* + 2y_2^* &= 2 \\
 y_1^* - y_2^* &< 3 \\
 2y_1^* + 3y_2^* &< 5 \\
 y_1^* + y_2^* &< 2 \\
 3y_1^* + y_2^* &= 3 \\
 y_1^* &> 0, y_2^* &> 0
 \end{aligned}$$

Using the complementary slackness condition to the above dual LP, we know that $x_2^* = 0, x_3^* = 0, x_4^* = 0$. Meanwhile, since $y_1^* = 0.8 > 0$ and $y_2^* = 0.6 > 0$, using the complementary slackness condition to

the primal LP, we have

$$x_1^* + x_2^* + 2x_3^* + x_4^* + 3x_5^* = 4 \quad (1)$$

$$2x_1^* - x_2^* + 3x_3^* + x_4^* + x_5^* = 3 \quad (2)$$

By setting $x_2^* = 0, x_3^* = 0, x_4^* = 0$ in (1)–(2), we obtain

$$x_1^* + 3x_5^* = 4$$

$$2x_1^* + x_5^* = 3$$

Therefore, the optimal primal solution is $x_1^* = 1, x_2^* = 0, x_3^* = 0, x_4^* = 0, x_5^* = 1$. The optimal objective function value for both primal and dual LPs is 5.

4. (10 points) PaGas is an energy corporation based in Pennsylvania. Currently, it has one fracking site, two gas extraction plants and a water recycling plant. The fracking site fractures the ground by pumping water, it requires 50 barrels (bbl) of water per day. The gas extraction plants generate water waste which cannot be disposed on the ground. This water must be either transported to the fracking site or to the water recycling plant. Currently, gas extraction plant A is generating 20 bbl/day of water waste while gas extraction plant B is generating 25 bbl/day. If the total amount of water transported to the fracking site is not enough to meet the daily demand of 50 bbl, then the company must purchase freshwater at a cost of \$70 per bbl in order to do so.

You are helping the manager of PaGas run their daily water operation. This involves deciding where to transport the water generated from each gas extraction plant (either to the fracking site or to the water recycling plant), and how much freshwater to purchase at the fracking site. The cost of transporting 1 bbl of water from the gas extraction plants A and B, to the fracking site and the water recycling plants can be found in the following table:

From Gas Extraction Plant	To Fracking Site	To Water Recycling Plant
A	\$200	\$100
B	\$180	\$120

Use the following decision variables:

- X_{AF} : Daily amount of water transported from gas extraction plant A to the fracking site.
- X_{BF} : Daily amount of water transported from gas extraction plant B to the fracking site.
- X_{AR} : Daily amount of water transported from gas extraction plant A to the water recycling plant.
- X_{BR} : Daily amount of water transported from gas extraction plant B to the water recycling plant.
- X_F : Daily amount of freshwater purchased at the fracking site.

Answer the following questions:

- (a) State a linear program (LP) that minimizes the cost of running the water operation for PaGas. Please make sure it has all required constraints.

$$\begin{aligned}
 \min \quad & 200X_{AF} + 100X_{AR} + 180X_{BF} + 120X_{BR} + 70X_F \\
 \text{s.t.} \quad & X_{AF} + X_{AR} = 20 \\
 & X_{BF} + X_{BR} = 25 \\
 & X_{AF} + X_{BF} + X_F = 50 \\
 & X_{AF}, X_{BF}, X_{AR}, X_{BR}, X_F \geq 0
 \end{aligned}$$

- (b) Solve the above LP using Excel (To obtain full points, you need to paste a picture of the spreadsheet showing the optimal solution you obtain from Excel).

The optimal solution is ($X_{AF}^* = 0, X_{AR}^* = 20, X_{BF}^* = 25, X_{BR}^* = 0, X_F^* = 25$)

5. (20 points) The Infinite Exploration Drift Airline needs to assign its crews to cover all its upcoming flights. The information about the flights and feasible sequences of flights for each crew is shown in the table below. For example, if you assign a crew to the last sequence of flights (i.e., the last column in the table), then the crew will cover the flights 3 (from San Francisco to Seattle), 4 (from Los Angeles to Chicago), 7 (from Chicago to Seattle), 10 (from Seattle to San Francisco), and 11 (from Seattle to Los Angeles).

Suppose there are three crews available, i.e., exactly three of the sequences of flights need to be chosen in such a way that every flight is covered. (It is permissible to have more than one crew on a flight, where the extra crews would fly as passengers, but union contracts require that the extra crews would still need to be paid for their time as if they were working). The cost of assigning a crew to a particular sequence of flights is given (in thousands of dollars) in the bottom row of the table. The objective is to minimize the total cost of the three crew assignments that cover all the flights. Formulate this problem as a binary integer program and solve it using Excel.

Flight	Feasible Sequence of Flights											
	1	2	3	4	5	6	7	8	9	10	11	12
1. San Francisco to Los Angeles	1			1			1			1		
2. San Francisco to Denver		1			1			1			1	
3. San Francisco to Seattle			1			1			1			1
4. Los Angeles to Chicago				2			2		3	2		3
5. Los Angeles to San Francisco	2					3				5	5	
6. Chicago to Denver				3	3				4			
7. Chicago to Seattle							3	3		3	3	4
8. Denver to San Francisco		2		4	4				5			
9. Denver to Chicago					2			2			2	
10. Seattle to San Francisco			2				4	4				5
11. Seattle to Los Angeles						2			2	4	4	2
Cost, \$1,000's	2	3	4	6	7	5	7	8	9	9	8	9

Solution: We use twelve binary decision variables to represent whether the sequence is selected. That is,

$$x_i = \begin{cases} 1, & \text{if sequence } i \text{ is assigned to a crew} \\ 0, & \text{otherwise,} \end{cases}$$

where $i = 1, 2, \dots, 12$. Then the objective function is defined as

$$\min \quad 2x_1 + 3x_2 + 4x_3 + 6x_4 + 7x_5 + 5x_6 + 7x_7 + 8x_8 + 9x_9 + 9x_{10} + 8x_{11} + 9x_{12}.$$

The constraints are

$$\begin{aligned}
x_1 + x_4 + x_7 + x_{10} &\geq 1 \\
x_2 + x_5 + x_8 + x_{11} &\geq 1 \\
x_3 + x_6 + x_9 + x_{12} &\geq 1 \\
x_4 + x_7 + x_9 + x_{10} + x_{12} &\geq 1 \\
x_1 + x_6 + x_{10} + x_{11} &\geq 1 \\
x_4 + x_5 + x_9 &\geq 1 \\
x_7 + x_8 + x_{10} + x_{11} + x_{12} &\geq 1 \\
x_2 + x_4 + x_5 + x_9 &\geq 1 \\
x_5 + x_8 + x_{11} &\geq 1 \\
x_3 + x_7 + x_8 + x_{12} &\geq 1 \\
x_6 + x_9 + x_{10} + x_{11} + x_{12} &\geq 1 \\
\sum_{i=1}^{12} x_i &= 3 \\
x_i &\in \{0, 1\}.
\end{aligned}$$

There are two optimal solutions for this problem. The first optimal solution is $(x_3^* = 1, x_4^* = 1, x_{11}^* = 1)$ and all other $x_i^* = 0$, for a total cost of \$18,000. The second optimal solution is $(x_1^* = 1, x_5^* = 1, x_{12}^* = 1)$, and all other $x_i^* = 0$.

6. (10 points) Suppose that we have an LP:

$$\begin{aligned}
\max_{x_1, x_2, x_3, x_4} \quad & x_1 + 2x_2 + 3x_3 + 4x_4 \\
\text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \leq 1.
\end{aligned}$$

We want to further include the following restrictions into this LP.

- (a) **At least one of the following two inequalities holds:**

$$\begin{aligned}
3x_1 + x_2 + x_3 + x_4 &\leq 12 \\
x_1 + x_2 + x_3 + x_4 &\leq 15
\end{aligned}$$

Solution:

$$\begin{aligned}
3x_1 + x_2 + x_3 + x_4 &\leq 12 + M(1 - y_1) \\
x_1 + x_2 + x_3 + x_4 &\leq 15 + M(1 - y_2) \\
y_1 + y_2 &\geq 1 \\
y &\in \{0, 1\}
\end{aligned}$$

- (b) **At least two of the following three inequalities holds:**

$$\begin{aligned}
2x_1 + 5x_2 + x_3 + x_4 &\leq 30 \\
x_1 + 3x_2 + 5x_3 + x_4 &\leq 40 \\
3x_1 + x_2 + 3x_3 + x_4 &\leq 60.
\end{aligned}$$

Solution:

$$\begin{aligned}
2x_1 + 5x_2 + x_3 + x_4 &\leq 30 + My_1 \\
x_1 + 3x_2 + 5x_3 + x_4 &\leq 40 + My_2 \\
3x_1 + x_2 + 3x_3 + x_4 &\leq 60 + My_3 \\
y_1 + y_2 + y_3 &\leq 1 \\
y_1, y_2, y_3 &\in \{0, 1\}
\end{aligned}$$

Or

$$\begin{aligned}2x_1 + 5x_2 + x_3 + x_4 &\leq 30 + M(1 - y_1) \\x_1 + 3x_2 + 5x_3 + x_4 &\leq 40 + M(1 - y_2) \\3x_1 + x_2 + 3x_3 + x_4 &\leq 60 + M(1 - y_3) \\y_1 + y_2 + y_3 &\geq 2 \\y_1, y_2, y_3 &\in \{0, 1\}\end{aligned}$$

Show how to reformulate these restrictions to fit an mixed integer program model.

7. (20 points) The Innovative Emerging and Development Analytics Company has been developing four possible new product lines. Management must now make a decision as to which of these four products actually will be produced and at what levels. Therefore, an operations research study has been requested to find the most profitable product mix.

A substantial cost is associated with beginning the production of any product, as given in the first row of the following table. Management's objective is to find the product mix that maximizes the total profit (total revenue minus start-up costs).

	Product			
	1	2	3	4
Start-up Cost	\$50,000	\$40,000	\$70,000	\$60,000
Marginal Revenue	\$70	\$60	\$90	\$80

Let the continuous decision variables x_1, x_2, x_3 , and x_4 be the production levels of products 1, 2, 3, and 4, respectively. Management has imposed the following policy constraints on these variables:

- No more than two of the products can be produced
- Either product 3 or 4 can be produced only if either product 1 or 2 is produced
- One of the two constraints holds:

$$\begin{aligned}5x_1 + 3x_2 + 6x_3 + 4x_4 &\leq 6,000 \\4x_1 + 6x_2 + 3x_3 + 5x_4 &\leq 6,000\end{aligned}$$

Answer the following questions:

- Introduce binary variables to formulate a mixed binary IP model for this problem.
- Use Excel to solve this model.

Solution: The math model is

$$\begin{aligned}\max \quad & 70x_1 - 50,000y_1 + 60x_2 - 40,000y_2 + 90x_3 - 70,000y_3 + 80x_4 - 60,000y_4 \\s.t. \quad & y_1 + y_2 + y_3 + y_4 \leq 2 \\& y_3 \leq y_1 + y_2 \\& y_4 \leq y_1 + y_2 \\& 5x_1 + 3x_2 + 6x_3 + 4x_4 \leq 6000 + My_5 \\& 4x_1 + 6x_2 + 3x_3 + 5x_4 \leq 6000 + M(1 - y_5) \\& 0 \leq x_i \leq My_i, \text{ for } i = 1, 2, 3, 4 \\& y_i \in \{0, 1\}, \text{ for } i = 1, 2, 3, 4\end{aligned}$$

Input the model into Excel and let M be a very large number, say 10^8 . The optimal solution obtained is $(x_2^* = 2000, y_2^* = 1)$. All other variables are 0. The maximized total profit is \$80,000.

8. (10 points) Suppose a discrete random variable is defined as follows:

$$X = \begin{cases} 0, & \text{with probability } 0.2 \\ 1, & \text{with probability } 0.5 \\ 2, & \text{with probability } 0.3 \end{cases}$$

What is $\mathbb{E}[X]$ and what is $\mathbb{E}[2X^2 + 3]$?

Solution: $\mathbb{E}[X] = 1.1$ and $\mathbb{E}[2X^2 + 3] = 6.4$